Identification-Based Simultaneous Reduction of Tangential and Normal Force Ripples in PMLSMs Under Dynamic Operation

Sangmin Lee, Yoon Sik Kwon, Jae Heon Jung, Kyujin Kang, Chang Hwan Kim, and Jun Young Yoon

***Abstract*—The iron-cored permanent-magnet linear synchronous motors (PMLSMs) are increasingly in demand with advantage of high force density and direct-drive capability. However, inherent force ripples caused by the magnetic saliency not only degrade the servo accuracy, but also induce vibration and noise, particularly during high-acceleration operations. This article presents a control-based force ripple reduction method to address such issues, by adjusting DQ-currents to counteract both tangential- and normal-direction force ripples simultaneously. Using an established motor force model, compensation DQ-currents are derived to generate additional forces that actively cancel out the force ripples. To enable effective compensation, experimental identification methods are developed to determine the force parameters required for the ripple reduction method, achieving a maximum identification error of less than 10 %. With these identified parameters, the proposed method experimentally achieves a reduction of over 87 % in the RMS value of tangential-direction force ripple and 69 % in the normal direction, all while maintaining constant motor thrust.**

***Index Terms*—DQ-currents, experimental identification, iron-cored linear synchronous motor, motor force ripple, reluctance force.**

I. Introduction

OWING to compactness, direct-drive capability, and high precision, permanent-magnet linear synchronous motors (PMLSMs) have received wide attention from many researchers [1], [2], [3]. Compared with the coreless type, iron-cored PMLSMs can generate higher thrust due to their shorter magnetic airgap. The iron-cores, however, introduce magnetic saliency in the airgap, which varies with mover position and causes force ripples in both the tangential (moving) and the normal directions [4], [5]. Additionally, excitation currents, which are applied to generate thrust, also contribute to force ripples [6]. Such force ripples not only impair the servo performance of PMLSMs, but also generate vibration and acoustic noise, which are critical issues in linear motor applications. Therefore, minimizing the force ripple in both tangential and normal directions is important. There are two main strategies for reducing force ripples: i) design optimization; and ii) control methods. While various design optimizations effectively mitigate parasitic force ripples [7], [8], [9], [10], these approaches are limited to the early design stages and usually complicate the manufacturing process. Thus, this article focuses on the control-based approach for force ripple reduction, which offers broader applicability.

In numerous prior studies, control methods have been investigated to reduce the force ripple, by generating additional motor forces to cancel it out. In [11], the injected currents are calculated to compensate for the detent force, but only a few dominant harmonics of the detent force are considered. In other studies [12] and [13], the force ripple is obtained using the spatial harmonic periods deduced by motor configuration, such as slotting and end effects, and corresponding harmonic amplitudes are online-estimated by adaptive control. In [14], the force ripple is also online-estimated based on the dynamics of a linear motor system, and suppressed using an adaptive feedforward compensator that generates a dither signal to counteract the force harmonics. However, these approaches focus on mitigating the geometry-driven ripple only, while neglecting the current-driven effects. To account for the current-driven effects, high order harmonics and the asymmetry of back-electromotive forces (bEMFs) are considered for current control strategy [15]. In [16], both the geometry-driven ripple and the current-driven ripple are taken into account, and experimental identification methods are also introduced. Despite the effectiveness of aforementioned studies in improving ripple reduction performance, they predominantly target the tangential-direction forces, leaving the normal direction force ripples unaddressed.

The significance of vibration and acoustic noise caused by normal-direction force ripples has been investigated in [17],

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[18], [19], where a double-sided mover configuration is proposed to mitigate such effects. Although these studies emphasize the importance of reducing the ripples, the limitations of the design approach still remain, such as challenging manufacturing process and higher cost. In [20], a novel commutation method for coreless PMLSMs is proposed to address normal-direction force ripples by analytically identifying them. However, this method is only validated in the finite element analysis (FEA) environment, without experimental validation. In [21], magnetic field in the airgap is measured using a hall sensor array and is employed to estimate force ripple coefficients, which are subsequently used to derive compensation control currents. This method is validated experimentally, but it is applied solely to ironless PMLSMs, where geometry-driven ripples from iron-cores are inherently absent and thus not considered. For iron-cored PMLSMs, our previous work in [22] experimentally identifies force coefficients and ripples, and compensates for tangential- and normal-direction force ripples by modeling the relationship between DQ-currents and forces in both directions. However, it is only effective for constant velocity regions as it neglects the reluctance forces, which become significant at high current levels due to its exponential relation to the input current [23]. This normal force increases as the PMLSM operates in high acceleration or for high load conditions, and induces the system vibration. However, such aspects have not been extensively explored in control-based ripple reduction studies.

In this article, we present the simultaneous force ripple reduction of iron-cored and surface-mounted PMLSMs in both tangential- and normal-directions. The proposed approach: 1) establishes a complete force model that incorporates both direction forces, explicitly considering the reluctance force during acceleration; 2) derives analytical compensating currents that account for cross-coupling between D- and Q-axis components; and 3) introduces experimental identification procedures for dominant parameters without the need for dedicated force sensors, enabling practical implementation. The proposed methods are experimentally validated using an iron-cored PMLSM with a 3-phase-4-magnet configuration, as schematically illustrated in Fig. 1. The rest of this article is organized as follows. Section II classifies the motor force types and derives the force equations. In Section III, the simultaneous ripple reduction method is presented, considering the coupling effects by D- and Q-axis currents and normal-direction reluctance force. Section IV introduces the experimental identification methods for force coefficients and force ripples in the normal direction. Section V and VI present the experimental testbed and the results validating the force identification and ripple reduction method. Section VII presents the simulation-based sensitivity analysis on force parameters, and then the article is concluded in Section VIII.

II. Motor Force Model

In this section, we present the motor force generation by D- and Q-axis currents in the normal and tangential directions. The magnetic force types are categorized based on the flux interaction types, and the relationship between D-Q axis currents

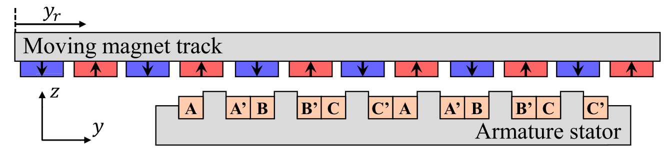

Fig. 1. Schematic diagram of an iron-cored PMLSM with a 3-phase, 4-magnet configuration.

and the resulting forces is analyzed. Subsequently, the motor force equations in both the normal and tangential directions are derived, which are later utilized in Section III for the force ripple reduction. Note that the iron-cored PMLSM is assumed to operate in the unsaturated region throughout this article, allowing the use of superposition principle in force modeling.

A. Motor Force Categorization

There are two types of magnetic flux in PMLSMs: PM- and electromagnet (EM)-driven flux. Depending on the magnetic flux types, we can define three types of magnetic forces: (1) geometry-driven force; (2) Lorentz force; and (3) reluctance force. When PM-driven flux engages solely with the armature iron, the geometry-driven force $F_{GDF}(y_r)$ is generated. Since the distance between PMs and iron-cores determines the force magnitude, the geometry-driven force depends on the mover position y_r. Note that $F_{GDF}(y_r)$ has dc attraction force of N along with ac force ripple in the normal direction since PMs tend to move closer to the armature iron, while the tangential geometry-driven force consists of only ac force ripple due to the symmetric configuration.

Lorentz force refers to the magnetic force resulting from the interaction between PM- and EM-driven fluxes. Since the D-axis current places the EM-driven flux beneath the PM poles, the Lorentz force is generated in the normal direction by attracting or repelling the PMs. Conversely, Q-axis current produces the EM-flux at the center of the alternating PMs, causing the Lorentz force to be generated in the tangential direction. Ideally, this means that the Lorentz force direction can be decoupled by D- and Q-axis currents. However, in practice, the realistic D- and Q-axis EM flux may deviate from their respective axes due to the unbalanced three-phase EM fluxes [24]. We therefore have a coupled Lorentz force in the normal direction by Q-axis current, and vice versa.

Reluctance force also arises when EM-driven flux solely engages with the mover yoke. This force is generated in the direction that minimizes magnetic reluctance, pulling the mover yoke toward the armature. Note that we in this article focus on the surface-mounted PM (SPM) type motors, where D- and Q-axis inductances are identical. As a result: (1) the reluctance force is assumed to act only in the normal direction; and (2) the force coefficients associated with the reluctance force are the same for both D- and Q-axis currents [25].

B. Motor Force Equation

Considering the categorized motor forces in the previous section, we derive the motor force equation in the tangential and normal directions, respectively, as

$$\begin{aligned}F_t(i_d, i_q, y_r) &= F_{GDF,t}(y_r) + F_{L,t}(i_d, i_q, y_r)\\ F_n(i_d, i_q, y_r) &= F_{GDF,n}(y_r) + F_{L,n}(i_d, i_q, y_r)\\ &\quad + F_{R,n}(i_d, i_q, y_r)\end{aligned} \tag{1}$$

where F_{GDF}, F_L, and F_R represent geometry-driven force, Lorentz force, and reluctance force, respectively. The subscripts t and n stand for the tangential and normal direction, respectively. The Lorentz force, which is linearly proportional to the D- and Q-axis currents [26], can be expressed as

$$\begin{aligned}F_{L,t}(i_d, i_q, y_r) &= \widetilde{k}_d(y_r) i_d + \left(\overline{k}_q + \widetilde{k}_q(y_r)\right) i_q\\ F_{L,n}(i_d, i_q, y_r) &= \left(\overline{h}_d + \widetilde{h}_d(y_r)\right) i_d + \widetilde{h}_q(y_r) i_q.\end{aligned} \tag{2}$$

The parameters k and h represent the force coefficients in the tangential and normal directions, respectively, while the subscripts d and q indicate the corresponding current axis. We use the tilde ($\widetilde{\ }$) to denote position-dependent ac fluctuating values and the overbar ($\overline{\ }$) to represent dc average values. Although the ac force terms in coupled directions are fully included in the above model, the ac terms in the normal direction, $\widetilde{h}_d$ and $\widetilde{h}_q$, are disregarded hereafter because: 1) their contribution is much smaller than the dominant dc terms [24]; and 2) the accurate experimental identification of position-dependent ac terms is challenging with the indirect measurement approach proposed in Section IV.

The reluctance force constant $\overline{h}_2$ is smaller than Lorentz force constants due to lower magnitude of the EM-driven flux density compared with the PM-driven flux density under air-cooled conditions [26]. For implementation in ripple reduction method, the reluctance force is simplified as: (1) the ac component of the reluctance force is neglected, given the small inductance variation in SPM-type motor; and (2) the reluctance force generated by D-axis current is ignored, as the D-axis current does not contribute to thrust generation, and thus remains significantly lower than the Q-axis current. As a result, we can express the reluctance force, which is quadratically proportional to the current [24], as

$$\begin{aligned}F_{R,n}(i_d, i_q, y_r) &= \overline{h}_2(i_d^2 + i_q^2)\\ &\simeq \overline{h}_2(i_q^2).\end{aligned} \tag{3}$$

Using the derived motor force equations, we compute the compensating D- and Q-axis current to simultaneously reduce the motor force ripples, as discussed in Section III, and present how the force coefficients are experimentally identified in Section IV.

III. Simultaneous Ripple Reduction Method in Both Tangential and Normal Directions

In this section, we present our method for simultaneously reducing force ripple in both tangential and normal directions, based on the force model discussed in Section II. The proposed method compensates not only the position-dependent ripple but also the current-driven reluctance force, which becomes particularly prominent during high-current operation, such as during acceleration phases. To achieve simultaneous suppression of all the undesired forces, both D- and Q-axis currents are simultaneously considered, and the compensating currents are calculated accordingly.

The desired motor forces in each direction are: (i) a constant thrust corresponding to the nominal Q-axis reference current $i_{q,\mathrm{ref}}$ in the tangential direction; and (ii) a constant attraction force N with a nominal D-axis current $i_{d,\mathrm{ref}} = 0\mathrm{A}$ in the normal direction, written as

$$\begin{aligned}F_{t,\mathrm{ref}}(i_d, i_q, y_r) &= \overline{k}_q i_{q,\mathrm{ref}}\\ F_{n,\mathrm{ref}}(i_d, i_q, y_r) &= N.\end{aligned} \tag{4}$$

We can begin with calculating the Q-axis current to reduce the tangential-direction force ripple by equating (1) and (4) as

$$i_{q,1} = \frac{-F_{GDF,t} + \overline{k}_q i_{q,\mathrm{ref}}}{k_q}. \tag{5}$$

To suppress the force ripple in the normal direction, the D-axis current compensating for both $\widetilde{F}_{GDF,n}(y_r)$, the ac component of $F_{GDF,n}(y_r)$, and the normal-direction force generated by $i_{q,1}$ can be expressed as

$$i_{d,1} = -\frac{\widetilde{F}_{GDF,n} + \overline{h}_2 i_{q,1}^2}{\overline{h}_d}. \tag{6}$$

Note that $i_{d,1}$ not only suppresses the force ripple in the normal direction, but also generates additional force ripple in the tangential direction simultaneously. Therefore, both compensating D- and Q-axis currents should be calculated iteratively, resulting in the following recurrence relation:

$$i_{q,n+1} = i_{q,1} - \frac{\widetilde{k}_d i_{d,n}}{k_q} \tag{7}$$

$$i_{d,n} = -\frac{\widetilde{F}_{GDF,n} + \overline{h}_2 i_{q,n}^2}{\overline{h}_d}. \tag{8}$$

By substituting (8) into (7), we rearrange the recurrence relation for compensating Q-axis current as

$$i_{q,n+1} = i_{q,1} + X(\widetilde{F}_{GDF,n} + \overline{h}_2 i_{q,n}^2) \tag{9}$$

where the coefficient $X(y_r)$ is defined as

$$X(y_r) = \frac{\widetilde{k}_d(y_r)}{k_q(y_r)\overline{h}_d}. \tag{10}$$

The final compensating Q-axis current is obtained by applying the contraction mapping theorem [27] to (9) as

$$i_{q,c} = \frac{1}{2X\overline{h}_2}\left(1 - \sqrt{1 - 4X\overline{h}_2(X\widetilde{F}_{GDF,n} + i_{q,1})}\right). \tag{11}$$

Subsequently, the final compensating D-axis current $i_{d,c}$ can be derived by taking the limit in (8) and then substituting (11) into (8).

From the derivation process, the compensating current $i_{q,c}$ should satisfy two conditions: (1) the convergence condition of the contraction mapping theorem; and (2) the requirement that it be a real value. First, the convergence condition of the

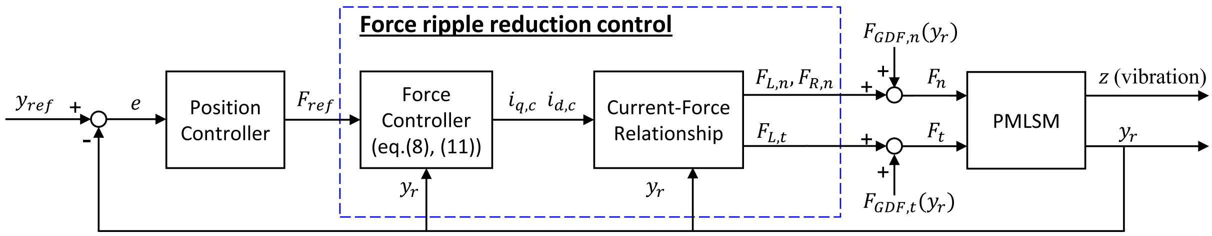

Fig. 2. Control block diagram using the proposed force ripple reduction method. The force control compensates for the position-dependent force ripple to generate accurate thrust as commanded by the position controller. The force ripples in the normal directions, which act as disturbance inducing the stage to vibrate, is mitigated simultaneously.

contraction mapping theorem is determined by the derivative of (9), given by

$$\left|\frac{\partial i_{q,n+1}}{\partial i_{q,n}} = 2X\bar{h}_2 i_{q,n}\right| < 1. \tag{12}$$

This can be satisfied by limiting the current amplitude as

$$|i_q| < i_{cont} = \frac{\min\left(|k_q||\bar{h}_d|\right)}{2\max\left(|\tilde{k}_d||\bar{h}_2|\right)} \le \left|\frac{k_q\bar{h}_d}{2\tilde{k}_d\bar{h}_2}\right|. \tag{13}$$

Note that the current limit i_{cont} is generally high, as the denominator includes only the coupled force ripple component and the reluctance force coefficient, both of which are relatively small compared with the dc values in the numerator. Second, the real value condition of (11) can be written as

$$X\bar{h}_2(X\tilde{F}_{GDF,n} + i_{q,1}) \le \frac{1}{4} \tag{14}$$

and it can also be satisfied by limiting amplitude of $i_{q,1}$ as

$$\begin{aligned} |i_{q,1}| &\le \frac{1}{4|X\bar{h}_2|} - |X\tilde{F}_{GDF,n}| \\ &= \frac{1}{2}i_{\text{cont}} - |X\tilde{F}_{GDF,n}| \end{aligned} \tag{15}$$

where i_{cont} is the maximum current limit for the contraction mapping theorem, expressed in (13). Note that $X\tilde{F}_{GDF,n}$ has a negligible amplitude, since $X(y_r)$ represents the ratio between force coefficients of the coupled and direct directions as expressed in (10). This leads to the maximum current limit being approximately $(1/2)i_{\text{cont}}$, which is stricter than the convergence condition of contraction mapping theorem.

By applying the compensating currents $i_{d,c}$ and $i_{q,c}$, the force ripples in both tangential and normal directions can be simultaneously reduced, leaving only the desired reference thrust F_{ref}. To implement this ripple reduction, a cascade control scheme is proposed, as shown in Fig. 2. The position controller generates a reference force $F_{\text{ref}} = \bar{k}_q i_{\text{ref}}$, which serves as the input to the force controller. The force controller then computes the compensating currents $i_{q,c}$ and $i_{d,c}$ to suppress position-dependent force disturbances $F_{GDF,t}$ and $\tilde{F}_{GDF,n}$. Note that the position controller can be designed based on a simplified PMLSM model that neglects force ripples, since the proposed ripple reduction control compensates for the position-dependent force ripple. The experimental validation of the proposed method is presented in Section VI.

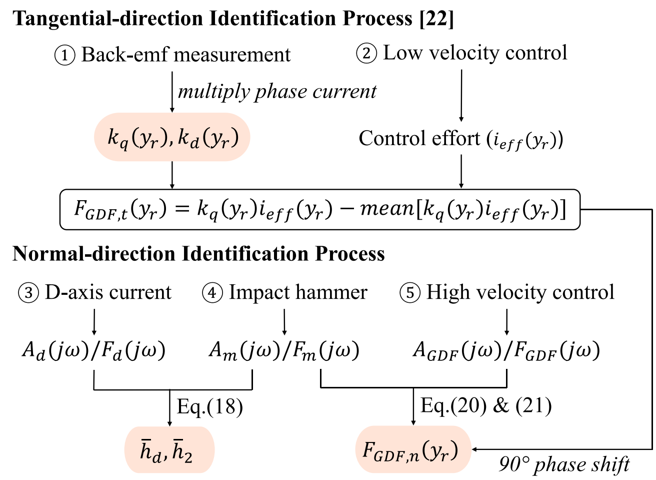

Fig. 3. Identification process of tangential- and normal-direction force parameters. The tangential-direction identification is detailed in [22].

IV. Experimental Force Identification Methods

This section presents the experimental methods for identifying the force parameters in the normal direction, which are used in the force ripple compensation presented in Section III. The force parameters in the tangential direction, k_q, k_d and $F_{GDF,t}$, can be directly identified by measuring the bEMFs and conducting constant velocity experiments. The detailed procedure for identifying these tangential-direction parameters is available in our previous work [22] and is not repeated here.

However, direct measurement of force parameters in the normal direction is not feasible due to the motion constraints imposed by the linear bearing system. Instead, indirect approaches using the vibration of the moving stage are employed to identify both the force coefficients and the geometry-driven ripple. Despite the limitations associated with indirect measurements, all dominant components of the force parameters are successfully identified, and subsequently used in the force ripple reduction method. The accuracy and validity of these identified parameters are demonstrated through experimental results presented in Section VI. The overall identification process for both tangential and normal force parameters is summarized in Fig. 3.

A. Force Coefficients in Normal Direction

For the force ripple reduction method described in Section III, it is necessary to identify the normal-direction force coefficients, $\overline{h}_d$ and $\overline{h}_2$. These coefficients correspond to the Lorentz force generated by d-axis current and the reluctance force generated by D- and Q-axis currents, respectively. Therefore, the D-axis current alone can generate both of these forces. To avoid any undesirable motion that might be induced by the Q-axis current during the identification process, only the D-axis current is utilized to identify $\overline{h}_d$ and $\overline{h}_2$.

These coefficients can be obtained by applying D-axis current inputs of varying magnitudes, and analyzing the resulting vibrations of the linear stage to quantify the induced current-driven forces. To measure the current-driven force, the impulse-like D-axis current is applied, which is expressed as

$$\begin{aligned} i_d(t) &= i_{\max}\delta_a(t) \\ &= i_{\max}\frac{1}{|a|\sqrt{\pi}}e^{-(t/a)^2} \end{aligned} \tag{16}$$

where $\delta_a(t)$ is a normalized Gaussian function, and a is a parameter that determines the temporal width of the distribution, effectively approximating a Dirac delta function as $a \to 0$. In this study, the value of a is set to 0.005 to provide a close approximation of an ideal impulse. The resulting current-driven force in the normal direction, which includes force terms that are linearly and quadratically proportional to the maximum current $i_{\max}$, is given by (3). This force can be expressed in the frequency domain by taking the Fourier transform as

$$F_d(j\omega) = \overline{h}_d i_{\max} D_1(j\omega) + \overline{h}_2 i_{\max}^2 D_2(j\omega) \tag{17}$$

where $D_1(j\omega)$ and $D_2(j\omega)$ are the Fourier transforms of $\delta_a(t)$ and $\delta_a^2(t)$, respectively. By measuring the acceleration caused by this current-driven force, the stage dynamics, H_{stage}, can be written as

$$H_{\text{stage}}(j\omega) = \frac{A_d(j\omega)}{F_d(j\omega)} = \frac{A_m(j\omega)}{F_m(j\omega)} \tag{18}$$

where A_d is the acceleration in the normal direction due to the current-driven force by D-axis current. The stage dynamics can also be obtained through impulse response analysis using an impact hammer, where F_m is the input force and A_m is the corresponding acceleration. Considering that the stage dynamics is identical for both force inputs, the current-driven force F_d can be calculated using (18).

By rearranging (17) and (18), the equation for $\overline{h}_d$ and $\overline{h}_2$ is derived as

$$\left|\frac{1}{H_{\text{stage}}(j\omega)}\frac{A_d(j\omega)}{i_{\max}D_1(j\omega)}\right| = \overline{h}_d + \overline{h}_2 i_{\max}\left|\frac{D_2(j\omega)}{D_1(j\omega)}\right|. \tag{19}$$

With $\overline{h}_d$ and $\overline{h}_2$ as the only unknowns, these force coefficients can be determined by comparing (19) for different magnitudes of $i_{\max}$. Fig. 4 schematically shows the identification process for force coefficients in the normal direction. The proposed identification method is experimentally verified and applied for force ripple reduction in Section VI.

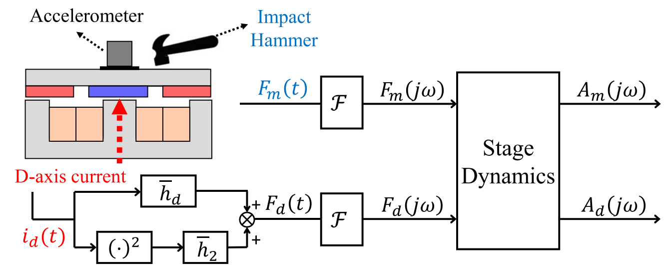

Fig. 4. Schematic diagram for the identification process of the normal direction force coefficients. The force coefficients can be identified by comparing the system responses, excited: i) mechanically by an impact hammer; and ii) electrically by the D-axis current.

B. Geometry-Driven Ripple in Normal Direction

The geometry-driven ripple in the normal direction can also be identified experimentally by focusing on the dominant harmonic components [22]. By decomposing it into several dominant harmonics, $\widetilde{F}_{GDF,n}(y_r)$ can be written as

$$\widetilde{F}_{GDF,n}(y_r) = \sum_l F_l \cos\left(\frac{2\pi}{\lambda_l}y_r + \phi_l\right) \tag{20}$$

where λ_l represents the spatial period of each harmonic, ϕ_l and F_l are the corresponding spatial phase and amplitude of the force ripple, respectively. Using this expression, the spatial period, phase and amplitude of each harmonic can be determined individually, and then combined to obtain the total force ripple.

The spatial periods and phases of the dominant harmonics in $\widetilde{F}_{GDF,n}(y_r)$ can be identified by referencing $F_{GDF,t}(y_r)$. Since the geometry-driven ripples in both directions originate from the same interaction between iron-cores and permanent magnets, they are expected to share the same dominant spatial period. Accordingly, the spatial period λ_l of $\widetilde{F}_{GDF,n}(y_r)$ is set equal to that of the dominant harmonic in $F_{GDF,t}(y_r)$. The corresponding spatial phase ϕ_l is determined from the inherent phase relationship between the two directions. The normal-direction force peaks when the magnets and cores are symmetrically aligned, which coincides with the zero-crossing point of the tangential-direction force. This results in a 90° phase difference between the two harmonics. Therefore, ϕ_l is obtained by shifting the phase of corresponding spatial harmonic in $F_{GDF,t}$ by 90° in electric angle.

Finally, the amplitude of the force harmonic is identified using the stage dynamics obtained in (18). Driving the linear motor at a constant velocity v_m, the harmonic of $\widetilde{F}_{GDF,n}$ acts as a sinusoidal force exerted in the normal direction, with its frequency ω_l being proportional to the velocity, $\omega_l = 2\pi v_m/\lambda_l$. Therefore, by measuring the acceleration during constant velocity motion and comparing it with the stage dynamics H_{stage}, the amplitude of the harmonic can be expressed as

$$F_l = |F_{GDF}(j\omega)|_{\omega=\omega_l} = \left|\frac{A_{GDF}(j\omega)}{H_{\text{stage}}(j\omega)}\right|_{\omega=\omega_l}. \tag{21}$$

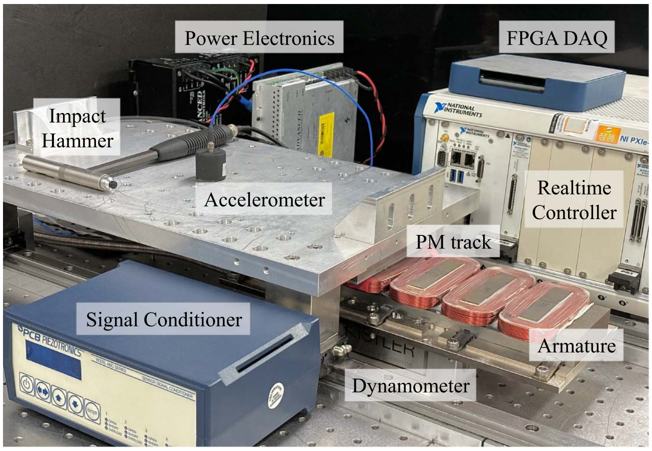

Fig. 5. Experimental testbed for validation of proposed identification and force ripple reduction methods. Linear stage with PM track is driven by the iron-cored armature stator, which is mounted on the dynamometer to directly measure the forces in both tangential and normal directions.

By combining the identified values, $\widetilde{F}_{GDF,n}$ can be reconstructed by (20). Note that this identification method is applicable to the force harmonics with different periods, as long as the resulting vibration is measurable.

V. Experimental Setup

Fig. 5 shows the experimental testbed of a linear motor stage, featuring a 3-phase-4-magnet combination PMLSM. The mover consists of eight poles of PMs (N38UH) with a pole pitch of 37.5 mm, while the armature stator has six iron-cored windings with a tooth pitch of 50 mm. The position of the mover is tracked by an optical encoder system by Renishaw (T1031-50A readhead, RTLC-S scale, and Ti0020A08A signal interface), with signal processed by the FPGA board (PXIe-7856R by National Instrument). This data is then sent to a real-time controller (PXIe-8861 by NI) operating at 10kHz, where the position control loop with force ripple compensation is implemented. The resulting current signals $i_{q,c}$ and $i_{d,c}$ are converted into three-phase currents, which are driven by a PWM amplifier (B30A40 by Advanced Motion Controls), powered by a dc power supply unit (PS30A by AMC).

A dynamometer (Type 9129AA by Kistler) is installed underneath the armature stator to measure the forces in both tangential and normal directions for validation purposes only. For force parameter identifications, an accelerometer (393B04 by PCB Piezotronics) is temporarily mounted on the moving stage, as discussed in Section IV. An impact hammer (086C03 by PCB) is employed together to measure the stage dynamics, which serves as a reference in the identification process. Note that the sensor resolution is sufficient to resolve the stage vibration, and the usable frequency range fully covers the bending mode of the stage, thereby ensuring reliable identification.

VI. Experimental Results

In this section, we present the experimental results of: (i) force parameter identification in the normal directions

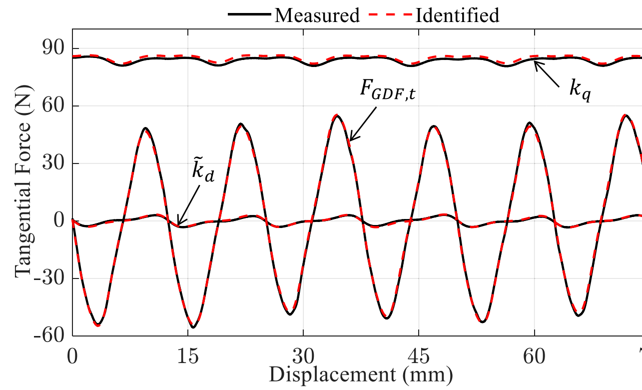

Fig. 6. Tangential-direction force parameters $k_q, \widetilde{k}_d$ and $F_{GDF,t}$, directly measured using a dynamometer (black line) and identified by proposed methods (red dashed line). Identification details are presented in [22].

TABLE I
Tangential-Direction Identification Results

Parameter	Identification Error Metric	Value (%)
$\overline{k}_q$	Mean-value error	1.39
$\widetilde{k}_q(y_r)$	Peak-to-peak normalized RMS error	5.30
$\widetilde{k}_d(y_r)$		4.70
$F_{GDF,t}(y_r)$		0.90

discussed in Section IV; and (ii) the proposed force ripple reduction method described in Section III. While the identification method is described for the normal direction in Section IV, the tangential-direction parameters, $k_q, \widetilde{k}_d$ and $F_{GDF,t}$, are also identified and compared with the measured values in Fig. 6 and Table I. A detailed discussion of the tangential direction has been reported in our previous work [22]. The validation of the proposed force ripple reduction method is conducted using the identified parameters, with particular emphasis on a high-acceleration trajectory where significant current-driven force ripples are produced.

A. Identification of Force Parameters

1) Force Coefficients: The force coefficients in the normal direction, $\overline{h}_d$ and $\overline{h}_2$, are identified by measuring the system responses of the stage, as discussed in Section IV-A. To obtain system responses, mechanical and electrical inputs are applied using an impact hammer and D-axis current, respectively, as shown in Fig. 7. First, the mechanical force is applied by the impact hammer, as shown by the black line in Fig. 7(a). Then, an impulse-like D-axis current is applied with a duration identical to that of the mechanical one. Since the normal-direction force by D-axis current consists of Lorentz and reluctance forces, which are proportional to $i_d(t)$ and $i_d^2(t)$, respectively, the electrical input is decomposed into these two components and represented separately by the green solid line and the blue dashed line in Fig. 7(b). By measuring the acceleration induced by these inputs, the frequency responses are obtained as shown in Fig. 8.

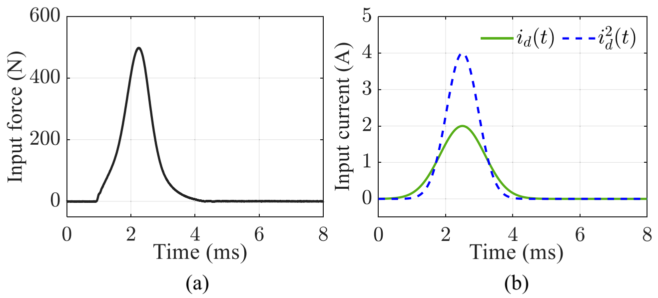

Fig. 7. Mechanical and electrical impulse inputs used to identify the normal-direction force coefficients. (a) Mechanical force input $F_m(t)$ excited by impact hammer, which directly excites the stage dynamics. (b) Electrical D-axis current input $i_d(t)$ and $i_d^2(t)$, which generate the Lorentz force ($F_{L,n}$) and the reluctance force ($F_{R,n}$), respectively.

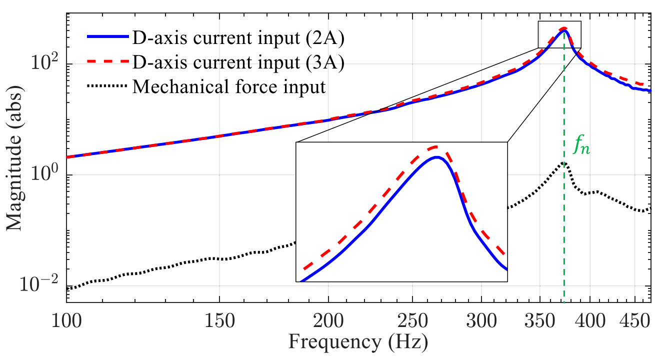

Fig. 8. Experimentally measured frequency responses of linear motor stage dynamics under mechanical and electrical excitations in the normal direction. Note that the responses to the electrical excitation vary with the input current level due to the reluctance force exhibiting a quadratic relation with the current.

The black dotted line in Fig. 8 represents the system response under mechanical excitation, which serves as a reference for the identification process. The blue solid and red dashed lines indicate the system responses under electrical excitation via D-axis currents with peak amplitudes of 2 A and 3 A, respectively, represented by $A_d(j\omega)/i_{\max}D_1(j\omega)$. A noticeable discrepancy between these electrically excited responses is observed in the magnified section of the plot. This deviation originates from the nonlinear relationship between the input current and the resulting normal-direction force output, as described in Section II.

Once these frequency responses are obtained, the two force coefficients are identified simultaneously by comparing the responses from mechanical and electrical excitations, using the identification formula in (19). The identified force coefficients, $\overline{h}_d$ and $\overline{h}_2$, are determined to be -296.7 N/A and -9.89 N/A^2, at the bending mode frequency of the stage, $f_n = 373$ Hz. These identified values show small errors of 0.57 % and 9.1 % compared with the measured values of -298.4 and -10.88 N/A^2. Note that $\overline{h}_2$ is significantly smaller than $\overline{h}_d$, supporting the assumption made in (3) that the reluctance force induced by the D-axis current is negligible.

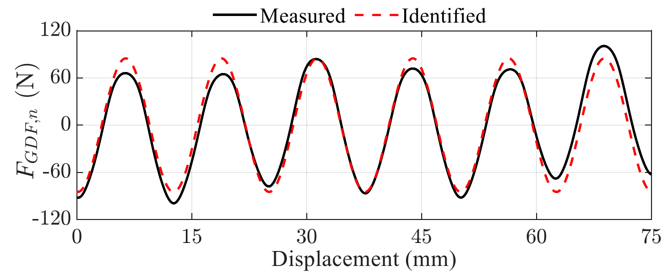

Fig. 9. Normal-direction geometry-driven force directly measured using a dynamometer (black line) and identified by proposed methods (red dashed line).

2) Geometry-Driven Forces: The geometry-driven force in the normal direction, $\widetilde{F}_{GDF,n}(y_r)$, is identified through a constant-velocity experiment. The motor is controlled at a high speed of 1.5 m/s, during which the stage vibration induced by $\widetilde{F}_{GDF,n}(y_r)$ is clearly observed. Among the harmonic components of $\widetilde{F}_{GDF,n}$, the most dominant one corresponds to the 6th-order harmonic of the pole-pair pitch, which has a spatial period of 12.5 mm. This harmonic induces a sinusoidal excitation of the stage at 120 Hz during the motion, causing it to vibrate at the same frequency. By comparing the vibration with the measured stage dynamics using (21), the harmonic amplitude is identified as 84.91 N, exhibiting an error of 4.25% compared with the measured value of 81.45 N. The phase of $\widetilde{F}_{GDF,n}$ is determined by shifting that of $F_{GDF,t}$ by 90 ° in electric angle, resulting in a spatial phase error of only 1.1 °. Using the identified amplitude and phase of the harmonic, the force ripple is reconstructed and compared with the measured one in Fig. 9, showing significant agreement. Despite considering only a single dominant harmonic, the identified force achieves an NRMSE (peak-to-peak normalized root-mean-square error) of 6.3% compared with the directly measured force, demonstrating the effectiveness of the proposed identification method.

B. Quasi-Static Measurement of Force Ripple Reduction

Fig. 10 presents the experimentally measured forces in both the tangential and normal directions under quasi-static conditions, where the PMLSM is slowly driven by an external feed-drive motor. The forces in each direction are compared under four conditions: (i) when the reference Q-axis current, $i_{q,ref}$, is set to 0 or 5 A; and (ii) with or without the ripple reduction method applied. Note that the reference current of 5 A represents the rated Q-axis current of the motor. The force ripples of each case are measured by a dynamometer in both directions, and are organized in Table II.

For the tangential direction shown in Fig. 10(a), the black dotted line represents uncompensated force ripple at $i_{q,\text{ref}} = 0$ A, which is identical to the geometry-driven force. As the ripple reduction method is implemented, the force ripple is significantly reduced by 96.2%, leaving force ripple of only 1.33 N in RMS, as shown by the blue solid line. For the case of $i_{q,\text{ref}} = 5$ A, the black and red solid lines represent the force ripple results without and with ripple reduction, respectively. The force ripple is also reduced by 87.5% when high level of

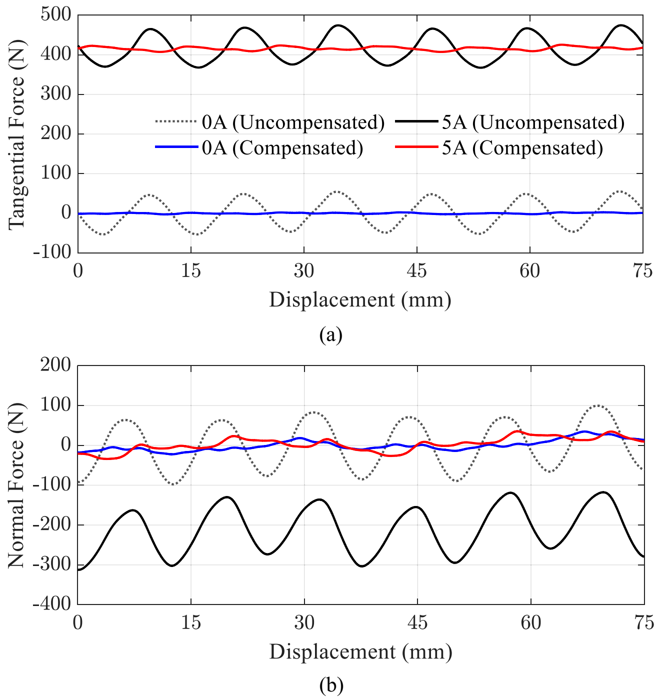

Fig. 10. Experimental results of force ripple reduction in tangential and normal directions, with reference Q-axis currents $i_{q,\text{ref}}$ of 0 and 5 A. (a) Tangential direction force ripples with and without the ripple reduction applied. The force ripples are significantly reduced while maintaining the average thrust. (b) Normal direction force ripples with and without ripple reduction applied. The force ripples and dc attraction force are reduced simultaneously using proposed method.

Q-axis current is applied, effectively suppressing both geometry-driven and current-driven ripples. Additionally, the dc component of the force, corresponding to the thrust generated by the Q-axis current, is consistently maintained at 416 N, with a small variation in thrust of less than 0.1%. These results demonstrate that the proposed method effectively reduces undesired ripple while maintaining the desired thrust, thereby enhancing servo performance as discussed in the following section.

On the other hand, the dc component of the normal-direction force should be eliminated, as it induces undesirable vibration rather than contributing to the intended motion along the moving direction. However, when the Q-axis current is applied to generate thrust, a reluctance force inevitably arises in the normal direction, as discussed in Section II. This reluctance force is illustrated by the black solid line in Fig. 10(b) for the case of $i_{q,\text{ref}} = 5$ A, indicating a dc force generation of -211.8 N compared with the 0 A case. With the proposed ripple reduction method applied, this dc force in the normal direction is significantly reduced to 3.07 N, achieving a 98.5% reduction in magnitude. At the same time, the ac components of the force ripple are also mitigated by 69.6%, as shown in Table II.

C. Dynamic Measurement of Force Ripple Reduction

The dynamic performance of the proposed ripple reduction method is evaluated during a trajectory tracking control. A high-acceleration trajectory is selected, where the parasitic forces in the normal direction become more pronounced due to the high Q-axis current for thrust generation. The trajectory follows a fourth-order polynomial curve with a maximum acceleration of 30 m/s^2, a maximum velocity of 0.4 m/s, and a travel distance of 75 mm. To evaluate the proposed method, a lead-lag compensator-based position controller with a crossover frequency of 60 Hz is applied identically to both cases, with and without the ripple reduction method.

TABLE II
FORCE RIPPLE REDUCTION RESULTS

$i_{q,\text{ref}}$	Direction	Force Ripple (RMS)	
		Without Reduction	With Reduction
0 A	Tangential	34.6 N	1.33 N (–96.2%)
	Normal	57.9 N	13.5 N (–79.7%)
5 A	Tangential	34.3 N	4.27 N (–87.5%)
	Normal	55.8 N	16.7 N (–69.6%)

Fig. 11(a) presents the position tracking errors with and without the proposed ripple reduction method, while Fig. 11(b) shows the corresponding tangential-direction force measured by a dynamometer. By preemptively compensating for the tangential-direction force ripple, which acts as a disturbance, the positioning error is significantly reduced in the constant velocity region, as highlighted in the magnified view. In particular, the 6th-order harmonic component of the positioning error, corresponding to the dominant harmonic in $F_{GDF,t}$, is reduced by 83.7%, from 50.8 μm to 8.2 μm. Note that although the force ripples are also suppressed during the acceleration phase, their influence is not clearly observable in either the positioning error or the measured tangential force. This is because the travel distance of acceleration region is too short to reveal position-dependent harmonics, and the force ripple magnitude is relatively small compared with the commanded thrust for acceleration.

In contrast to the tangential direction, the normal-direction force ripple is clearly observable and significantly reduced in both the acceleration and constant velocity phases, as shown in Fig. 11(c). This improvement is attributed to the ripple reduction algorithm, which compensates for the normal-direction attraction force by treating it as a disturbance during acceleration, while preserving the commanded thrust in the tangential direction. By suppressing both geometry-driven and current-driven forces, the peak-to-peak ripple in the normal direction is reduced by 83.2%, from 793.7 N to 133.5 N. Note that the high-frequency components observed in both Fig. 11(b) and (c) originate from the bending modes of the stage, which are beyond the frequency range targeted by the proposed ripple reduction method.

Fig. 11(d) shows the corresponding DQ-current profiles when the ripple reduction method is applied. During the acceleration phase, the Q-axis current peaks at 8.57 A, which is slightly higher than 8.47 A observed without the ripple reduction method, in order to counteract the geometry-driven ripple in advance. This high Q-axis current, if left uncompensated, generates a substantial normal-direction attraction force

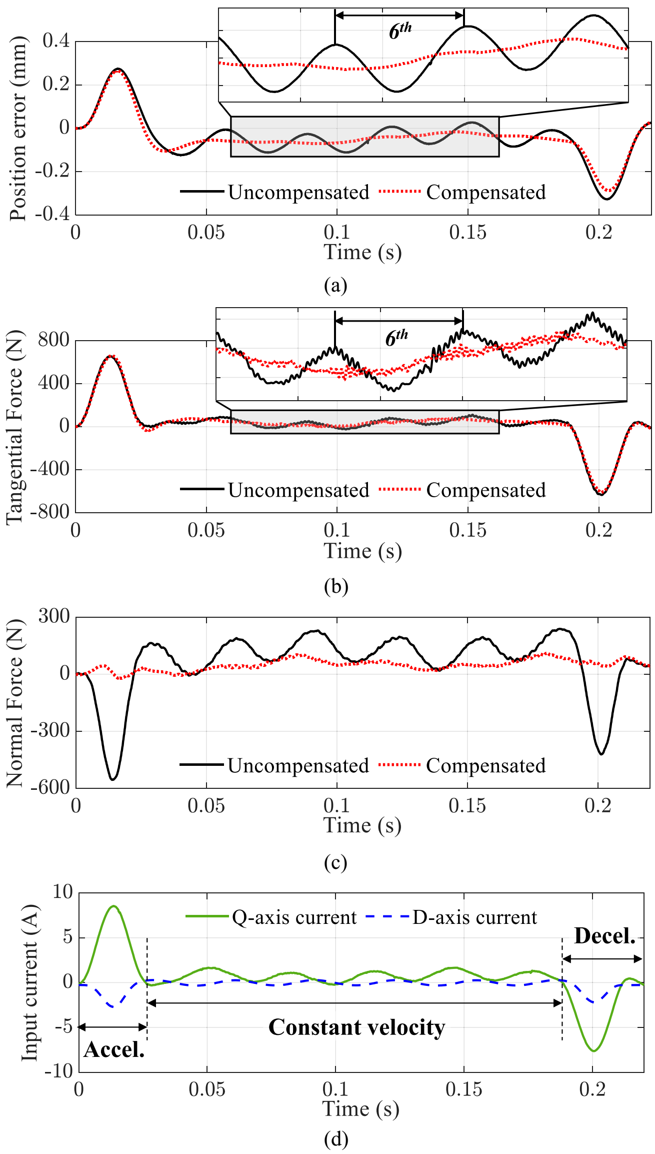

Fig. 11. Experimental results of trajectory tracking control. (a) Position tracking error; (b) measured tangential-direction; (c) normal-direction force with and without the ripple reduction method applied; and (d) D- and Q-axis current profiles when the ripple reduction is applied. High level of Q-axis currents in acceleration and deceleration regions induce reluctance force, which is compensated by D-axis current.

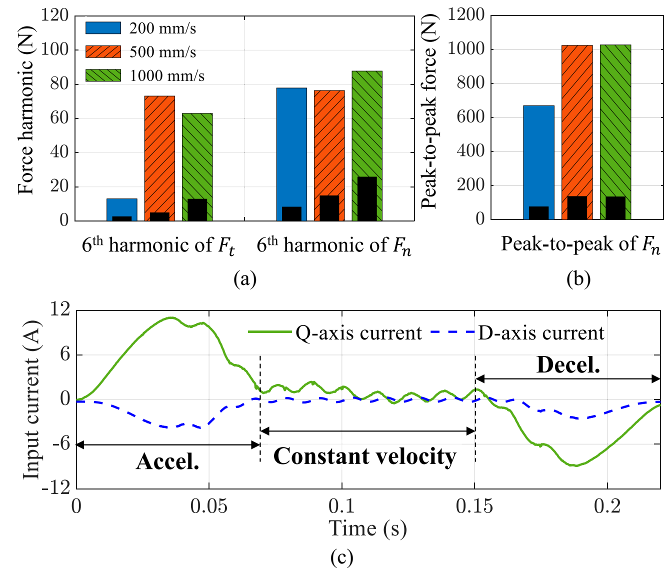

Fig. 12. Comparison of the remaining force with and without force ripple compensation at various speeds. (a) Dominant 6th harmonic measured in the tangential (F_t) and normal directions (F_n) during the constant velocity region. (b) Peak-to-peak of F_n measured during acceleration, mainly consisting of the reluctance force. Colored hatched bars indicate uncompensated results, and black bars indicate corresponding compensated results. (c) D- and Q-axis current profiles for the operation at 1,000 mm/s with ripple reduction applied.

reaching up to −555.2 N. To mitigate this attraction force, the D-axis current is simultaneously applied, reaching a peak value of −2.73 A. During the constant velocity phase, the D-axis current remains below 0.34 A in magnitude, mainly addressing the geometry-driven ripple. Note that, despite the additional use of D-axis current, the total RMS current is only slightly increased from 2.58 A without the ripple reduction method to 2.63 A with it, indicating efficient current allocation with minimal additional consumption.

To further validate the proposed method in various operating conditions, including higher current, experimental results at different speeds and with an additional load of 12.5 kg are presented in Fig. 12. The motor is driven at 200, 500, and 1000 mm/s, with the same maximum acceleration of 25 m/s^2. The colored hatched bars in Fig. 12 represent the uncompensated results at different speeds, while the black bars represent corresponding compensated results. Fig. 12(a) shows the most dominant force component, the 6th-order harmonic in both tangential and normal directions, is compared before and after compensation. In Fig. 12(b), for acceleration phase, only the peak-to-peak force in the normal direction is evaluated, since the tangential force is directly used for acceleration and is not a reliable measure of ripple suppression in this regime.

The results show that the 6th-order harmonics are reduced by more than 70% in both directions across all tested speeds. The reduction rate slightly decreases as the motor speed becomes higher due to the combined delay of the digital control loop and the current driver. In the acceleration phase, the normal-direction peak-to-peak force, mainly consisting of the reluctance force by accelerating Q-axis current, is reduced by more than 85% at every speed using the proposed method. Note that the lowest peak is observed at 200 mm/s, because the maximum acceleration is not fully reached under the feedback control. These results confirm that the proposed method effectively suppresses force ripple under both constant velocity and high acceleration conditions, demonstrating its robustness for a wide range of speeds and load conditions typical in industrial transportation applications.

Fig. 12(c) shows the DQ-current profiles for the 1000 mm/s operation, which represents a high-current condition. Under this operation, the Q-axis current reaches a peak of 11.04 A during acceleration, generating a substantial reluctance force in the normal direction exceeding 1000 N. To counteract this

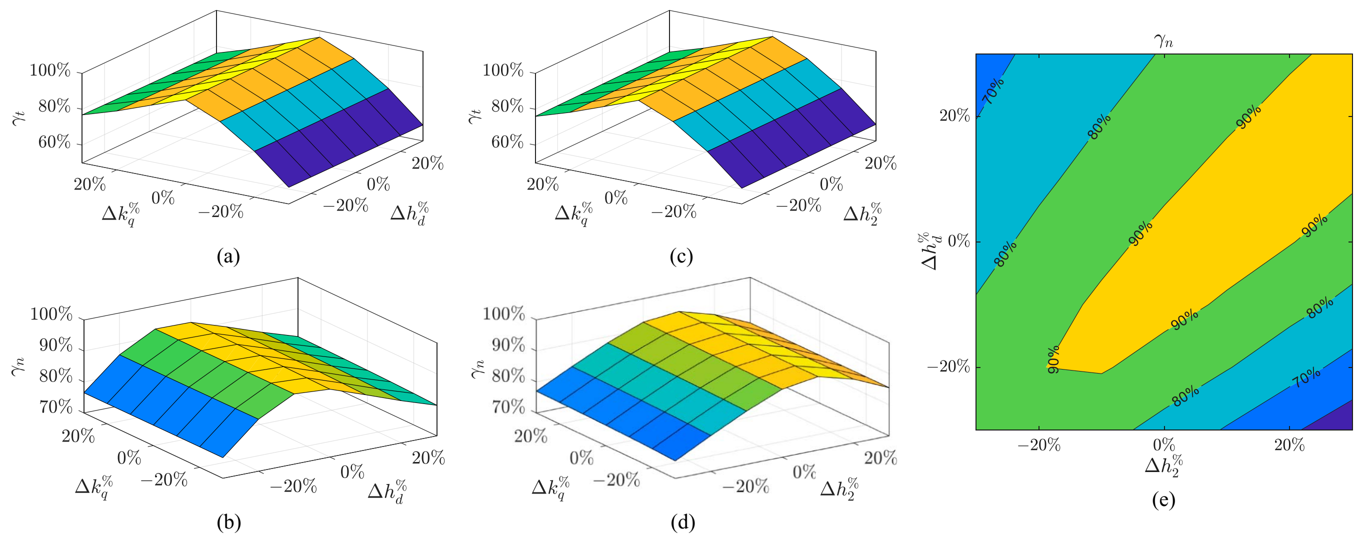

Fig. 13. Reduction ratios of residual forces according to parameter variations. (a) Tangential- and (b) normal-direction reduction ratios with respect to normalized variations of $\Delta k_q^{\%}$ and $\Delta h_d^{\%}$. (c) Tangential- and (d) normal-direction reduction ratios for $\Delta k_q^{\%}$ and $\Delta h_2^{\%}$. (e) Normal-direction reduction ratio for $\Delta h_d^{\%}$ and $\Delta h_2^{\%}$.

attraction force, the D-axis current is simultaneously applied, reaching a peak of 3.83 A. During the constant velocity phase, both D- and Q-axis currents exhibit a dominant 80 Hz harmonic component, corresponding to the 6th-order spatial harmonics of $F_{GDF,t}$ and $F_{GDF,n}$. These harmonic currents significantly suppress the geometry-driven force ripples in both tangential and normal directions, as shown in Fig. 12(a). Similar to the low-current case in Fig. 11, the total RMS current during this high-current operation increases only slightly from 5.29 A without compensation to 5.44 A with compensation. This demonstrates that the proposed method achieves efficient current utilization and robust performance even under high-load, high-speed operation.

VII. Sensitivity Analysis

The actual forces of PMLSMs may deviate from the force model calculated with parameters employed in the compensation, due to identification errors, magnetic saturation, or temperature variations. To validate the effectiveness and robustness of the proposed method under such practical conditions, this section investigates the effect of variations in the dominant parameters, $\overline{k}_q$, $\overline{h}_d$, and $\overline{h}_2$, on force ripple compensation. Each parameter is perturbed within $\pm 30\%$ of its nominal value, covering both typical identification errors and possible deviations under operating conditions. The analysis is conducted using finite element simulations with a reference Q-axis current of 5 A, where two parameters are simultaneously varied while the remaining one is fixed at its nominal value.

To evaluate the compensation performance, the reduction ratios of the force ripples are defined separately for tangential and normal directions as

$$\gamma_t = 1 - \frac{\max(F_{t,\mathrm{comp}}) - \min(F_{t,\mathrm{comp}})}{\max(F_{t,\mathrm{uncomp}}) - \min(F_{t,\mathrm{uncomp}})} \tag{22}$$

$$\gamma_n = 1 - \frac{\mathrm{rms}(F_{n,\mathrm{comp}} - N)}{\mathrm{rms}(F_{n,\mathrm{uncomp}} - N)} \tag{23}$$

where N denotes the nominal attraction force used in (4), calculated as the average normal force under zero-current excitation. Note that (23) is introduced to simultaneously capture the reduction of ac ripple components and the dc attraction force caused by the reluctance force.

Fig. 13 shows the reduction ratios with respect to normalized parameter deviations of $\Delta k_q^{\%}$, $\Delta h_d^{\%}$, and $\Delta h_2^{\%}$, as written in

$$\Delta k_q^{\%} = \frac{\overline{k}_q - \overline{k}_{q,\mathrm{nom}}}{\overline{k}_{q,\mathrm{nom}}} \times 100 \tag{24}$$

$$\Delta h_d^{\%} = \frac{\overline{h}_d - \overline{h}_{d,\mathrm{nom}}}{\overline{h}_{d,\mathrm{nom}}} \times 100 \tag{25}$$

$$\Delta h_2^{\%} = \frac{\overline{h}_2 - \overline{h}_{2,\mathrm{nom}}}{\overline{h}_{2,\mathrm{nom}}} \times 100. \tag{26}$$

Fig. 13(a) and (b) present the case where $\overline{k}_q$ and $\overline{h}_d$ are varied while $\overline{h}_2$ is fixed. Here, the reduction ratio in tangential direction, γ_t decreases as $\overline{k}_q$ deviates from the nominal value, while it remains almost consistent for $\overline{h}_d$, reflecting the relatively small contribution of the D-axis current to tangential force generation. In contrast, the normal-direction ratio γ_n is highly affected by variations in $\overline{h}_d$, while it is barely influenced by $\overline{k}_q$. This is because a deviation in k_q alters the Q-axis current from its ideal value, but the resulting reluctance force is still accurately predicted and canceled out by the D-axis current.

Fig. 13(c) and (d) show the results when $\overline{k}_q$ and $\overline{h}_2$ are varied with $\overline{h}_d$ fixed. Similarly, $\overline{k}_q$ predominantly governs the tangential-direction compensation, whereas $\overline{h}_2$ governs the normal-direction performance. Within the examined range shown in Fig. 13(a)–(d), γ_t and γ_n remain higher than 58% and 75%, respectively, demonstrating substantial suppression compared with the uncompensated case.

Fig. 13(e) presents the normal-direction reduction ratio when $\overline{h}_d$ and $\overline{h}_2$ are varied while $\overline{k}_q$ is fixed. Note that Fig. 13(e) focuses on the normal-direction response, because the tangential-direction ratio γ_t remains nearly constant. As both $\overline{h}_d$ and $\overline{h}_2$ correspond to the normal direction, γ_n shows a strong correlation with their simultaneous variation. When both parameters increase or decrease together, the Lorentz force modeled by $\overline{h}_d$ continues to counteract the reluctance force by $\overline{h}_2$, preserving reduction performance over 90%. In contrast, when one parameter increases while the other decreases, the imbalance between the Lorentz and reluctance forces is amplified, leading to deterioration in performance. Nevertheless, even under these unfavorable conditions, γ_n remains above 50%, confirming the robustness and effectiveness of the proposed compensation method.

VIII. Conclusion

This article presents an improved control method for suppressing force ripples in both tangential and normal directions of PMLSMs. The electromagnetic forces are classified and modeled based on their relationship with input currents, enabling the derivation of compensating DQ-currents. The compensating currents are then used to generate additional forces that cancel out the fluctuating force ripples simultaneously in both directions. Experimental methods for identifying normal-direction force parameters are also proposed, allowing practical implementation without the need for permanently mounted equipment. The identification method is experimentally validated showing a maximum error of less than 10%. The ripple reduction method, implemented using the identified parameters, is also experimentally validated with a significant reduction rate of over 87% and 69% in the tangential and normal directions, respectively. The servo tracking control experiments with and without the proposed ripple reduction also illustrate that incorporating the reluctance force into the reduction method actively suppress the force ripples during acceleration. In addition, a simulation-based sensitivity analysis on the dominant parameters was performed, confirming that the proposed method maintains robust and effective ripple suppression under practical operating conditions. These results demonstrate significant potential for the proposed method in applying high-force PMLSMs for applications that demand both high acceleration and high precision.

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